

House Price Prediction in King County

Regression, PCA, and Clustering Analysis via KNIME

Machine Learning Project — University of Milano-Bicocca

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Abstract

This report presents a machine learning analysis of the King County House Sales dataset (21,612 residential transactions, 2014–2015). We implement a complete KNIME workflow covering data preprocessing, feature engineering, four regression models (Linear Regression, Decision Tree, Random Forest, Gradient Boosted Trees), dimensionality reduction via PCA, 5-fold cross-validation, and K-Means clustering. Random Forest achieves the best predictive performance (Test $R^2 = 0.703$, RMSE = 0.292 on log-price). PCA with 12 components (95% variance) yields $R^2 = 0.654$. Clustering identifies two distinct market segments with a silhouette score of 0.248. Feature importance analysis confirms that living space, construction grade, and neighbourhood quality are the primary determinants of house prices.

Keywords: House Price Prediction, Random Forest, PCA, K-Means Clustering, KNIME, King County

Introduction

Accurate residential property valuation is of fundamental importance to buyers, sellers, financial institutions, and urban planners. The availability of large transaction datasets, combined with modern machine learning methods, enables data-driven price prediction at scale. This study analyses the King County House Sales dataset from Kaggle, which records 21,612 property sales in King County, Washington State (USA) between May 2014 and May 2015.

Our research question is: *Which features most strongly determine house prices in King County, and how accurately can machine learning models predict them?* To answer this, we compare four regression algorithms, apply PCA for dimensionality reduction, validate all models via 5-fold cross-validation, and segment the market using K-Means clustering. The entire pipeline is implemented in KNIME Analytics Platform.

Dataset and Preprocessing

Dataset Description

The raw dataset contains 21,613 rows and 21 columns. The target variable is **price** (continuous, range: \$75,000–\$7,700,000). After removing a data entry error (one house with 33 bedrooms in 1,620 sqft), 21,612 records remain.

Feature Engineering and Preprocessing

The preprocessing pipeline consists of the following steps:

1. **Column removal:** `id`, `date`, `zipcode` removed (non-informative).
2. **Feature engineering:** `house_age` = $2015 - \text{yr_built}$; binary `renovated` (1 if `yr_renovated` > 0); binary `basement` (1 if `sqft_basement` > 0).
3. **Redundant column removal:** `yr_built`, `yr_renovated`, `sqft_basement`, `sqft_above` excluded to avoid multicollinearity.
4. **Missing value imputation:** median strategy (no missing values found in dataset).
5. **Log transformation:** `log_price` = $\ln(\text{price})$ to reduce right skew and improve linearity.

6. **Normalisation:** Z-score normalisation fitted on training set only, applied to test set via Normalizer Apply node to prevent data leakage.
7. **Train/test split:** 70% training (15,128 rows), 30% test (6,484 rows), random seed = 42.

After preprocessing, **14 features** are retained: `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, `floors`, `waterfront`, `view`, `condition`, `grade`, `sqft_living15`, `sqft_lot15`, `house_age`, `renovated`, `basement`.

Correlation Analysis

Table 1 shows the strongest pairwise correlations. `sqft_living` and `grade` are most correlated with `log_price`. High mutual correlations between size features confirm the importance of living space and quality.

Table 1: Selected pairwise Pearson correlations ($|\rho| > 0.44$)

Feature pair	Pearson ρ	Interpretation
<code>sqft_living</code> – <code>grade</code>	0.763	Size correlates with quality
<code>sqft_living</code> – <code>sqft_living15</code>	0.756	Neighbourhood size effect
<code>sqft_living</code> – <code>bathrooms</code>	0.755	Larger homes have more bathrooms
<code>grade</code> – <code>bathrooms</code>	0.665	Quality and amenities linked
<code>bathrooms</code> – <code>bedrooms</code>	0.529	Room count correlation
<code>floors</code> – <code>house_age</code>	−0.489	Newer homes tend to be multi-floor
<code>grade</code> – <code>house_age</code>	−0.447	Newer homes built to higher grade

Methods

Regression Models

Four regression models are trained to predict `log_price`:

- **Linear Regression (LR):** Ordinary least squares baseline.
- **Decision Tree (DT):** CART algorithm, max depth = 8, min records per node = 5.
- **Random Forest (RF):** Ensemble of 100 trees, random seed = 42, square-root feature sampling.
- **Gradient Boosted Trees (GBT):** 200 trees, learning rate = 0.05, max depth = 6, row sampling = 0.8.

Dimensionality Reduction via PCA

PCA is applied to the 14 normalised training features (excluding `price` and `log_price`) using KNIME’s PCA Compute node. Components are selected by retaining 95% of total variance, yielding **12 principal components**. A Linear Regression model is then trained on the PCA-transformed features and evaluated on the PCA-projected test set.

5-Fold Cross-Validation

Independent 5-fold CV loops are implemented for each model (LR, DT, RF, GBT) using separate X-Partitioner/X-Aggregator pairs, connected to the full preprocessed dataset before the train/test split. Linear sampling and seed = 42 ensure reproducibility.

K-Means Clustering

K-Means is applied to 10 selected features (`sqft_living`, `grade`, `lat`, `long`, `price`, `bathrooms`, `view`, `waterfront`, `house_age`, `renovated`) after Z-score normalisation on the full dataset. $K \in \{2, 3, 4, 5\}$ is evaluated via the Silhouette Coefficient to select the optimal number of clusters.

Results

Regression — Holdout Evaluation

Table 2 reports test set performance. **Random Forest** achieves the best R^2 and RMSE. All metrics refer to the log-price scale.

Table 2: Holdout test set performance (target: `log_price`)

Model	Test R^2	Test RMSE	Test MAE	Adj. R^2
Random Forest	0.703	0.292	0.231	0.703
Gradient Boosted Trees	0.696	0.295	0.231	0.696
Linear Regression	0.663	0.311	0.247	0.663
LR + PCA (12 dims)	0.654	0.315	0.229	0.653
Decision Tree	0.620	0.330	0.260	0.620

Key observations: (i) RF outperforms all models; (ii) GBT ranks second, confirming the strength of boosting; (iii) LR+PCA ($R^2=0.654$) falls below full LR (0.663), showing the information cost of dimensionality reduction; (iv) DT shows the weakest performance, indicating overfitting at depth 8.

5-Fold Cross-Validation

Table 3 reports average MSE and RMSE across folds. Rankings are consistent with holdout results, confirming stability.

Table 3: 5-fold cross-validation results (average across folds)

Model	Avg. MSE	Avg. RMSE	Rank
Random Forest	0.0820	0.286	1 st
Gradient Boosted Trees	0.0874	0.296	2 nd
Linear Regression	0.0949	0.308	3 rd
Decision Tree	0.1070	0.327	4 th

CV confirms RF as the best model across all splits. DT’s drop in CV rank (4th vs. moderate holdout) indicates sensitivity to the specific split. Low fold-to-fold variance confirms stable generalisation for all models.

Price Distribution and Log Transformation

Figure 1 illustrates the effect of the log transformation. The raw price distribution is strongly right-skewed; after log transformation, the distribution becomes approximately normal, satisfying regression assumptions.

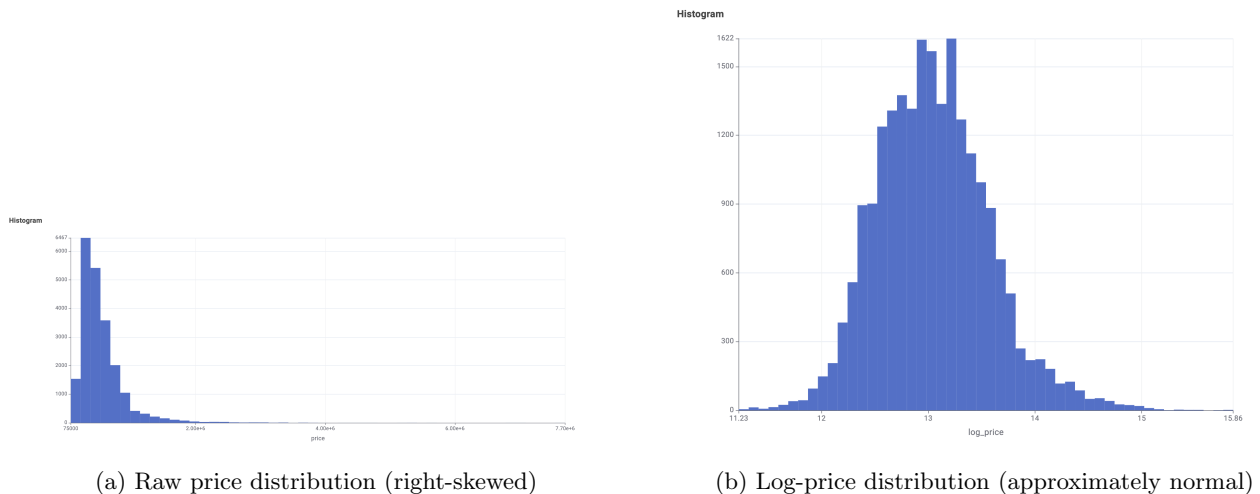
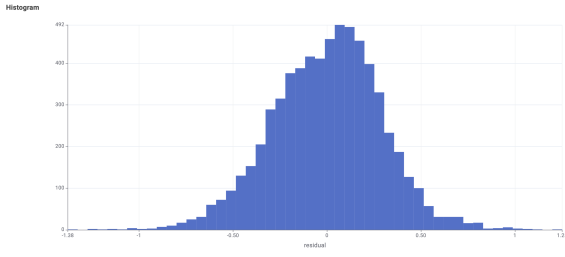


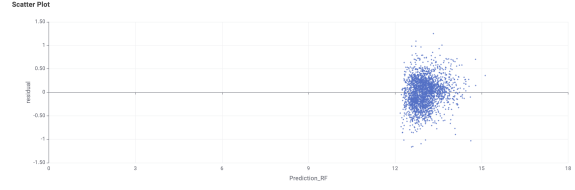
Figure 1: Effect of log transformation on the target variable distribution.

Residual Analysis

Figure 2 shows the residual analysis for Random Forest. The histogram exhibits a bell-curve shape centred at zero, and the scatter plot shows no systematic pattern, confirming that the model’s errors are approximately normally distributed and unbiased.



(a) Residual histogram (bell curve, centred at 0)



(b) Residuals vs. predicted values (random scatter)

Figure 2: Residual analysis for Random Forest on the holdout test set.

K-Means Clustering

Table 4 shows Silhouette scores for $K = 2, 3, 4, 5$. Scores decrease monotonically with K ; $K = 2$ is selected as optimal.

Table 4: Silhouette scores for $K = 2, 3, 4, 5$

K	Overall Silhouette
2	0.248
3	0.217
4	0.199
5	0.182

With $K = 2$, two distinct market segments emerge (Figure 3):

- **Cluster 0 — Standard market:** 13,961 properties (65%) — below-average price, smaller living space, lower construction grade, older houses.
- **Cluster 1 — Premium market:** 7,651 properties (35%) — above-average price (+0.70 SD), larger living space (+0.94 SD), higher grade (+0.95 SD), higher waterfront proportion.

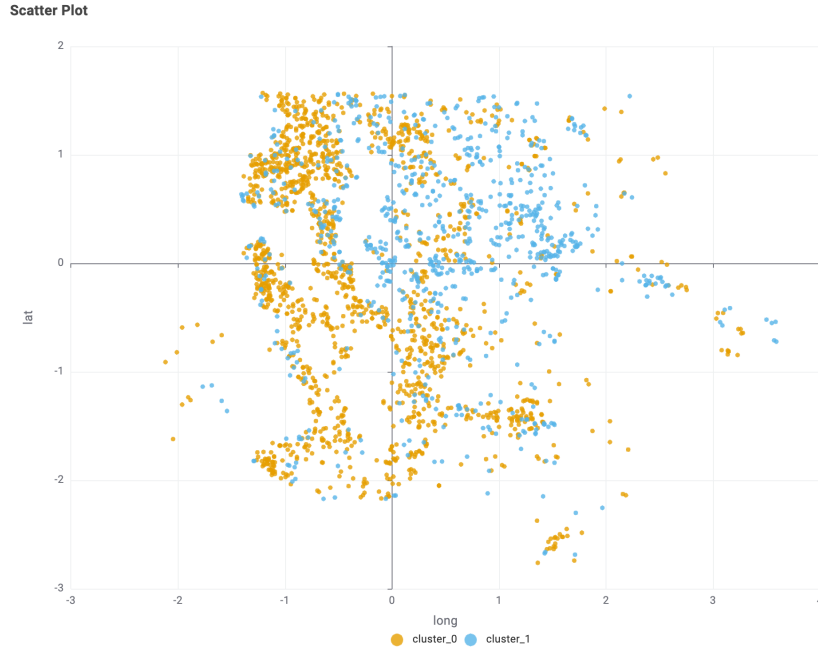


Figure 3: Geographic distribution of K-Means clusters ($K = 2$) on normalised lat/long coordinates. Two price segments are clearly visible across King County.

Feature Importance

Random Forest attribute statistics (splits at tree level 0) identify the most discriminative features (Table 5). Living space, amenity count, and construction quality emerge as the top predictors, consistent with the correlation analysis.

Table 5: Top features by Random Forest splits at level 0

Feature	Splits (level 0)	Role
bathrooms	20	Amenity count
sqft_living15	19	Neighbourhood living area
sqft_living	16	Interior size
grade	13	Construction quality
floors	11	Building structure
view	5	Aesthetic value

Conclusions

This study demonstrates that machine learning models can predict King County house prices accurately from structural and locational features. **Random Forest** achieves the best performance ($R^2 = 0.703$, RMSE = 0.292), confirmed by 5-fold cross-validation. **Gradient Boosted Trees** provides a competitive alternative ($R^2 = 0.696$). Linear Regression offers a robust interpretable baseline ($R^2 = 0.663$).

PCA with 12 components (95% variance) achieves $R^2 = 0.654$, demonstrating effective dimensionality reduction at a modest predictive cost. K-Means clustering ($K = 2$, Silhouette = 0.248) reveals two natural market segments — standard and premium — with clear geographic separation across King County.

Feature importance consistently highlights **living space** (`sqft_living`, `sqft_living15`), **construction quality** (`grade`), and **amenities** (`bathrooms`) as primary price drivers. These findings provide actionable insights for real estate valuation and urban planning.

Limitations: The dataset covers one year (2014–2015) and one region. Models do not capture temporal dynamics or macroeconomic factors. Future work could incorporate time-series modelling, spatial autocorrelation, and external economic indicators.

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